

NAME: HAFREEN ALI

ROLL NO: 22L-6298

SECTION: 6A

COURSE: DIGITAL SIGNAL PROCESSING

ASSIGNMENT #2

$$1) a) x_p(n) = \cos\left(\frac{2\pi}{10}n\right), \quad -\infty < n < \infty$$

$$f_0 = \frac{1}{10}, \quad x(n) = x_p(n), \quad 0 \leq n \leq N-1, \quad N=10$$

$$\therefore \cos x = \frac{e^{jx} + e^{-jx}}{2}$$

$$\Rightarrow x(n) = \frac{1}{2} \left[e^{j\left(\frac{2\pi}{10}n\right)} + e^{-j\left(\frac{2\pi}{10}n\right)} \right]$$

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi}{N}kn\right)} \quad \therefore N=10$$

$$\Rightarrow X(k) = \sum_{n=0}^{N-1} \left[\left(\frac{1}{2} e^{j\left(\frac{2\pi}{10}n\right)} \right) \left(e^{-j\left(\frac{2\pi}{10}kn\right)} \right) + \left(\frac{1}{2} e^{-j\left(\frac{2\pi}{10}n\right)} \right) \left(e^{-j\left(\frac{2\pi}{10}kn\right)} \right) \right]$$

$$X(k) = \sum_{n=0}^{N-1} \left[\frac{1}{2} e^{-j\left(\frac{2\pi}{10}\right)(k-1)n} \right] + \sum_{n=0}^{N-1} \left[\frac{1}{2} e^{-j\left(\frac{2\pi}{10}\right)(k+1)n} \right] \quad (i)$$

For $k=0$, $n=0, 1, 2, \dots, 9$

$$\begin{aligned} \Rightarrow (i) \quad X(0) &= \frac{1}{2} \left[e^{-j\left(\frac{2\pi}{10}\right)(-1)(0)} + e^{-j\left(\frac{2\pi}{10}\right)(+1)(0)} \right] + \frac{1}{2} \left[e^{-j\left(\frac{2\pi}{10}\right)(-1)(1)} + e^{-j\left(\frac{2\pi}{10}\right)(+1)(1)} \right] + \\ &\quad \frac{1}{2} \left[e^{-j\left(\frac{2\pi}{10}\right)(-1)(2)} + e^{-j\left(\frac{2\pi}{10}\right)(+1)(2)} \right] + \frac{1}{2} \left[e^{-j\left(\frac{2\pi}{10}\right)(-1)(3)} + e^{-j\left(\frac{2\pi}{10}\right)(+1)(3)} \right] + \frac{1}{2} \left[e^{-j\left(\frac{2\pi}{10}\right)(-1)(4)} + \right. \\ &\quad \left. e^{-j\left(\frac{2\pi}{10}\right)(+1)(4)} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times (-1) \times 5} + e^{-j\frac{2\pi}{10} \times (+1) \times 5} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times (-1) \times 6} + e^{-j\frac{2\pi}{10} \times (+1) \times 6} \right] + \\ &\quad \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times (-1) \times 7} + e^{-j\frac{2\pi}{10} \times (+1) \times 7} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times (-1) \times 8} + e^{-j\frac{2\pi}{10} \times (+1) \times 8} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times (-1) \times 9} + e^{-j\frac{2\pi}{10} \times (+1) \times 9} \right] \end{aligned}$$

$$\begin{aligned} X(0) &= \frac{1}{2} \left[e^0 + e^0 \right] + \frac{1}{2} \left[e^{j\frac{\pi}{5}} + e^{-j\frac{\pi}{5}} \right] + \frac{1}{2} \left[e^{j\frac{2\pi}{5}} + e^{-j\frac{2\pi}{5}} \right] + \frac{1}{2} \left[e^{j\frac{3\pi}{5}} + e^{-j\frac{3\pi}{5}} \right] + \\ &\quad \frac{1}{2} \left[e^{j\frac{4\pi}{5}} + e^{-j\frac{4\pi}{5}} \right] + \frac{1}{2} \left[e^{j\frac{5\pi}{5}} + e^{-j\frac{5\pi}{5}} \right] + \frac{1}{2} \left[e^{j\frac{6\pi}{5}} + e^{-j\frac{6\pi}{5}} \right] + \frac{1}{2} \left[e^{j\frac{7\pi}{5}} + e^{-j\frac{7\pi}{5}} \right] \end{aligned}$$

$$\frac{1}{2} \left[e^{j\frac{8\pi}{5}} + e^{-j\frac{8\pi}{5}} \right] + \frac{1}{2} \left[e^{j\frac{9\pi}{5}} + e^{-j\frac{9\pi}{5}} \right]$$

$$\begin{aligned} X(0) = & \frac{1}{2}(2) + \frac{1}{2} \left[\cos \frac{\pi}{5} + j \sin \frac{\pi}{5} + \cos \frac{\pi}{5} - j \sin \frac{\pi}{5} \right] + \frac{1}{2} \left[\cos \frac{2\pi}{5} + j \sin \frac{2\pi}{5} + \cos \frac{2\pi}{5} - j \sin \frac{2\pi}{5} \right] \\ & + \frac{1}{2} \left[\cos \frac{3\pi}{5} + j \sin \frac{3\pi}{5} + \cos \frac{3\pi}{5} - j \sin \frac{3\pi}{5} \right] + \frac{1}{2} \left[\cos \frac{4\pi}{5} + j \sin \frac{4\pi}{5} + \cos \frac{4\pi}{5} - j \sin \frac{4\pi}{5} \right] \\ & + \frac{1}{2} \left[\cos \frac{5\pi}{5} + j \sin \frac{5\pi}{5} + \cos \frac{5\pi}{5} - j \sin \frac{5\pi}{5} \right] + \frac{1}{2} \left[\cos \frac{6\pi}{5} + j \sin \frac{6\pi}{5} + \cos \frac{6\pi}{5} - j \sin \frac{6\pi}{5} \right] \\ & + \frac{1}{2} \left[\cos \frac{7\pi}{5} + j \sin \frac{7\pi}{5} + \cos \frac{7\pi}{5} - j \sin \frac{7\pi}{5} \right] + \frac{1}{2} \left[\cos \frac{8\pi}{5} + j \sin \frac{8\pi}{5} + \cos \frac{8\pi}{5} - j \sin \frac{8\pi}{5} \right] \\ & + \frac{1}{2} \left[\cos \frac{9\pi}{5} + j \sin \frac{9\pi}{5} + \cos \frac{9\pi}{5} - j \sin \frac{9\pi}{5} \right] \end{aligned}$$

$$X(0) = 0$$

$$\text{For } k=1, n=0, 1, 2, \dots, 9$$

$$\begin{aligned} k-1 &= 1-0=1 \\ k+1 &= 1+1=2 \end{aligned}$$

$$\begin{aligned} X(1) = & \frac{1}{2} \left[e^{-j\left(\frac{2\pi}{10}\right)(0)(0)} + e^{-j\frac{2\pi}{10} \times 2 \times 0} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 0 \times 1} + e^{-j\frac{2\pi}{10} \times 2 \times 1} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 0 \times 2} + e^{-j\frac{2\pi}{10} \times 2 \times 2} \right] \\ & + \frac{1}{2} \left[e^{-j\left(\frac{2\pi}{10}\right) \times 0 \times 3} + e^{-j\frac{2\pi}{10} \times 2 \times 3} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 0 \times 4} + e^{-j\frac{2\pi}{10} \times 2 \times 4} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 0 \times 5} + e^{-j\frac{2\pi}{10} \times 2 \times 5} \right] \\ & + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 0 \times 6} + e^{-j\frac{2\pi}{10} \times 2 \times 6} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 0 \times 7} + e^{-j\frac{2\pi}{10} \times 2 \times 7} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 0 \times 8} + e^{-j\frac{2\pi}{10} \times 2 \times 8} \right] \\ & + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 0 \times 9} + e^{-j\frac{2\pi}{10} \times 2 \times 9} \right] \end{aligned}$$

$$\begin{aligned} X(1) = & \frac{1}{2} \left[e^0 + e^0 \right] + \frac{1}{2} \left[e^0 + e^{-j\frac{4\pi}{10}} \right] + \frac{1}{2} \left[e^0 + e^{-j\frac{8\pi}{10}} \right] + \frac{1}{2} \left[e^0 + e^{-j\frac{12\pi}{10}} \right] + \frac{1}{2} \left[e^0 + e^{-j\frac{16\pi}{10}} \right] \\ & + \frac{1}{2} \left[e^0 + e^{-j\frac{20\pi}{10}} \right] + \frac{1}{2} \left[e^0 + e^{-j\frac{24\pi}{10}} \right] + \frac{1}{2} \left[e^0 + e^{-j\frac{28\pi}{10}} \right] + \frac{1}{2} \left[e^0 + e^{-j\frac{32\pi}{10}} \right] + \frac{1}{2} \left[e^0 + e^{-j\frac{36\pi}{10}} \right] \end{aligned}$$

$$\begin{aligned} X(1) = & 1 + \frac{1}{2} \left[1 + \cos \frac{4\pi}{10} - j \sin \frac{4\pi}{10} \right] + \frac{1}{2} \left[1 + \cos \frac{8\pi}{10} - j \sin \frac{8\pi}{10} \right] + \frac{1}{2} \left[1 + \cos \frac{12\pi}{10} - j \sin \frac{12\pi}{10} \right] \\ & + \frac{1}{2} \left[1 + \cos \frac{16\pi}{10} - j \sin \frac{16\pi}{10} \right] + \frac{1}{2} \left[1 + \cos \frac{20\pi}{10} - j \sin \frac{20\pi}{10} \right] + \frac{1}{2} \left[1 + \cos \frac{24\pi}{10} - j \sin \frac{24\pi}{10} \right] \\ & + \frac{1}{2} \left[1 + \cos \frac{28\pi}{10} - j \sin \frac{28\pi}{10} \right] + \frac{1}{2} \left[1 + \cos \frac{32\pi}{10} - j \sin \frac{32\pi}{10} \right] + \frac{1}{2} \left[1 + \cos \frac{36\pi}{10} - j \sin \frac{36\pi}{10} \right] \end{aligned}$$

$$X(1) = 5$$

For $k=2$, $n=0, 1, 2, \dots, 9$

$$k-1 = 2-1 = 1$$

$$k+1 = 2+1 = 3$$

(i)

$$X(2) = \left[\frac{1}{2} e^{-j\frac{2\pi}{10} \times 1 \times 0} + \frac{1}{2} e^{-j\frac{2\pi}{10} \times 3 \times 0} \right] + \left[\frac{1}{2} e^{-j\frac{2\pi}{10} \times 1 \times 1} + \frac{1}{2} e^{-j\frac{2\pi}{10} \times 3 \times 1} \right] + \left[\frac{1}{2} e^{-j\frac{2\pi}{10} \times 1 \times 2} + \frac{1}{2} e^{-j\frac{2\pi}{10} \times 3 \times 2} \right] +$$

$$\left[\frac{1}{2} e^{-j\frac{2\pi}{10} \times 1 \times 3} + \frac{1}{2} e^{-j\frac{2\pi}{10} \times 3 \times 3} \right] + \left[\frac{1}{2} e^{-j\frac{2\pi}{10} \times 1 \times 4} + \frac{1}{2} e^{-j\frac{2\pi}{10} \times 3 \times 4} \right] + \left[\frac{1}{2} e^{-j\frac{2\pi}{10} \times 1 \times 5} + \frac{1}{2} e^{-j\frac{2\pi}{10} \times 3 \times 5} \right] +$$

$$\left[\frac{1}{2} e^{-j\frac{2\pi}{10} \times 1 \times 6} + \frac{1}{2} e^{-j\frac{2\pi}{10} \times 3 \times 6} \right] + \left[\frac{1}{2} e^{-j\frac{2\pi}{10} \times 1 \times 7} + \frac{1}{2} e^{-j\frac{2\pi}{10} \times 3 \times 7} \right] + \left[\frac{1}{2} e^{-j\frac{2\pi}{10} \times 1 \times 8} + \frac{1}{2} e^{-j\frac{2\pi}{10} \times 3 \times 8} \right] +$$

$$\left[\frac{1}{2} e^{-j\frac{2\pi}{10} \times 1 \times 9} + \frac{1}{2} e^{-j\frac{2\pi}{10} \times 3 \times 9} \right]$$

$$X(2) = \frac{1}{2} [e^0 + e^0] + \frac{1}{2} \left[\cos \frac{2\pi}{10} - j \sin \frac{2\pi}{10} + \cos \frac{6\pi}{10} - j \sin \frac{6\pi}{10} \right] + \frac{1}{2} \left[\cos \frac{4\pi}{10} - j \sin \frac{4\pi}{10} \right]$$

$$+ \cos \frac{12\pi}{10} - j \sin \frac{12\pi}{10} \Big] + \frac{1}{2} \left[\cos \frac{6\pi}{10} - j \sin \frac{6\pi}{10} + \cos \frac{18\pi}{10} - j \sin \frac{18\pi}{10} \right] +$$

$$\frac{1}{2} \left[\cos \frac{8\pi}{10} - j \sin \frac{8\pi}{10} + \cos \frac{24\pi}{10} - j \sin \frac{24\pi}{10} \right] + \frac{1}{2} \left[\cos \frac{10\pi}{10} - j \sin \frac{10\pi}{10} + \cos \frac{30\pi}{10} - j \sin \frac{30\pi}{10} \right]$$

$$+ \frac{1}{2} \left[\cos \frac{12\pi}{10} - j \sin \frac{12\pi}{10} + \cos \frac{36\pi}{10} - j \sin \frac{36\pi}{10} \right] + \frac{1}{2} \left[\cos \frac{14\pi}{10} - j \sin \frac{14\pi}{10} + \cos \frac{42\pi}{10} - j \sin \frac{42\pi}{10} \right]$$

$$+ \frac{1}{2} \left[\cos \frac{16\pi}{10} - j \sin \frac{16\pi}{10} + \cos \frac{48\pi}{10} - j \sin \frac{48\pi}{10} \right] + \frac{1}{2} \left[\cos \frac{18\pi}{10} - j \sin \frac{18\pi}{10} + \cos \frac{54\pi}{10} - j \sin \frac{54\pi}{10} \right]$$

$$X(2) = 1 + \frac{1}{2} [0.809 - j0.5877 - 0.309 - j0.951] + \frac{1}{2} [0.309 - j0.951 - 0.809 + j0.5877] +$$

$$\frac{1}{2} [-0.309 - j0.951 + j0.5877 + 0.809] + \frac{1}{2} [-0.809 - j0.5877 - j0.951 + 0.309] +$$

$$\frac{1}{2} [-1 - j0 - j0 - 1] + \frac{1}{2} [-0.809 + j0.5877 + j0.951 + 0.309] + \frac{1}{2} [-0.309 + j0.951 - j0.5877 + 0.809]$$

$$+ \frac{1}{2} [0.309 + j0.951 - j0.5877 - 0.809] + \frac{1}{2} [0.809 + j0.5877 + j0.951 - 0.309]$$

$$X(2) = 0$$

For $k=5$, $n=0,1,2,\dots,9$

$k-1 = 5-1 = 4$
 $k+1 = 5+1 = 6$

(i)

$$X(5) = \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 4 \times 0} + e^{-j\frac{2\pi}{10} \times 6 \times 0} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 4 \times 1} + e^{-j\frac{2\pi}{10} \times 6 \times 1} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 4 \times 2} + e^{-j\frac{2\pi}{10} \times 6 \times 2} \right] +$$

$$\frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 4 \times 3} + e^{-j\frac{2\pi}{10} \times 6 \times 3} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 4 \times 4} + e^{-j\frac{2\pi}{10} \times 6 \times 4} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 4 \times 5} + e^{-j\frac{2\pi}{10} \times 6 \times 5} \right] +$$

$$\frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 4 \times 6} + e^{-j\frac{2\pi}{10} \times 6 \times 6} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 4 \times 7} + e^{-j\frac{2\pi}{10} \times 6 \times 7} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 4 \times 8} + e^{-j\frac{2\pi}{10} \times 6 \times 8} \right] +$$

$$\frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 4 \times 9} + e^{-j\frac{2\pi}{10} \times 6 \times 9} \right]$$

$$X(5) = \frac{1}{2} [e^0 + e^0] + \frac{1}{2} \left[\cos \frac{8\pi}{10} - j \sin \frac{8\pi}{10} + \cos \frac{12\pi}{10} - j \sin \frac{12\pi}{10} \right] + \frac{1}{2} \left[\cos \frac{16\pi}{10} - j \sin \frac{16\pi}{10} + \right.$$

$$\left. \cos \frac{24\pi}{10} - j \sin \frac{24\pi}{10} \right] + \frac{1}{2} \left[\cos \frac{24\pi}{10} - j \sin \frac{24\pi}{10} + \cos \frac{36\pi}{10} - j \sin \frac{36\pi}{10} \right] +$$

$$\frac{1}{2} \left[\cos \frac{32\pi}{10} - j \sin \frac{32\pi}{10} + \cos \frac{48\pi}{10} - j \sin \frac{48\pi}{10} \right] + \frac{1}{2} \left[\cos \frac{40\pi}{10} - j \sin \frac{40\pi}{10} + \right.$$

$$\left. \cos \frac{60\pi}{10} - j \sin \frac{60\pi}{10} \right] + \frac{1}{2} \left[\cos \frac{48\pi}{10} - j \sin \frac{48\pi}{10} + \cos \frac{72\pi}{10} - j \sin \frac{72\pi}{10} \right] +$$

$$\frac{1}{2} \left[\cos \frac{56\pi}{10} - j \sin \frac{56\pi}{10} + \cos \frac{84\pi}{10} - j \sin \frac{84\pi}{10} \right] + \frac{1}{2} \left[\cos \frac{64\pi}{10} - j \sin \frac{64\pi}{10} + \right.$$

$$\left. \cos \frac{96\pi}{10} - j \sin \frac{96\pi}{10} \right] + \frac{1}{2} \left[\cos \frac{72\pi}{10} - j \sin \frac{72\pi}{10} + \cos \frac{108\pi}{10} - j \sin \frac{108\pi}{10} \right]$$

$$X(5) = 0$$

For $k=7$, $n=0,1,2,\dots,9$

$k-1 = 7-1 = 6$
 $k+1 = 7+1 = 8$

(i)

$$X(7) = \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 6 \times 0} + e^{-j\frac{2\pi}{10} \times 8 \times 0} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 6 \times 1} + e^{-j\frac{2\pi}{10} \times 8 \times 1} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 6 \times 2} + e^{-j\frac{2\pi}{10} \times 8 \times 2} \right] +$$

$$\frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 6 \times 3} + e^{-j\frac{2\pi}{10} \times 8 \times 3} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 6 \times 4} + e^{-j\frac{2\pi}{10} \times 8 \times 4} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 6 \times 5} + e^{-j\frac{2\pi}{10} \times 8 \times 5} \right] +$$

$$\frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 6 \times 6} + e^{-j\frac{2\pi}{10} \times 8 \times 6} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 6 \times 7} + e^{-j\frac{2\pi}{10} \times 8 \times 7} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 6 \times 8} + e^{-j\frac{2\pi}{10} \times 8 \times 8} \right] +$$

$$\frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 6 \times 9} + e^{-j\frac{2\pi}{10} \times 8 \times 9} \right]$$

$$\begin{aligned}
 X(7) = & \frac{1}{2} [e^0 + e^0] + \frac{1}{2} \left[\cos \frac{12\pi}{10} - j \sin \frac{12\pi}{10} + \cos \frac{16\pi}{10} - j \sin \frac{16\pi}{10} \right] + \frac{1}{2} \left[\cos \frac{24\pi}{10} - \right. \\
 & j \sin \frac{24\pi}{10} + \cos \frac{32\pi}{10} - j \sin \frac{32\pi}{10} \left. \right] + \frac{1}{2} \left[\cos \frac{36\pi}{10} - j \sin \frac{36\pi}{10} + \cos \frac{48\pi}{10} - j \sin \frac{48\pi}{10} \right. \\
 & + \frac{1}{2} \left[\cos \frac{48\pi}{10} - j \sin \frac{48\pi}{10} + \cos \frac{64\pi}{10} - j \sin \frac{64\pi}{10} \right] + \frac{1}{2} \left[\cos \frac{60\pi}{10} - j \sin \frac{60\pi}{10} + \cos \frac{80\pi}{10} \right. \\
 & - j \sin \frac{80\pi}{10} \left. \right] + \frac{1}{2} \left[\cos \frac{72\pi}{10} - j \sin \frac{72\pi}{10} + \cos \frac{96\pi}{10} - j \sin \frac{96\pi}{10} \right] + \frac{1}{2} \left[\cos \frac{84\pi}{10} - j \sin \frac{84\pi}{10} \right. \\
 & \cos \frac{112\pi}{10} - j \sin \frac{112\pi}{10} \left. \right] + \frac{1}{2} \left[\cos \frac{96\pi}{10} - j \sin \frac{96\pi}{10} + \cos \frac{128\pi}{10} - j \sin \frac{128\pi}{10} \right] + \\
 & \frac{1}{2} \left[\cos \frac{108\pi}{10} - j \sin \frac{108\pi}{10} + \cos \frac{144\pi}{10} - j \sin \frac{144\pi}{10} \right]
 \end{aligned}$$

$$X(7) = 0$$

For $k=9$, $n=0, 1, 2, \dots, 9$

$$k-1 = 9-1 = 8$$

$$k+1 = 9+1 = 10$$

$$\begin{aligned}
 \Rightarrow (i) \\
 X(9) = & \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 8 \times 0} + e^{-j\frac{2\pi}{10} \times 10 \times 0} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 8 \times 1} + e^{-j\frac{2\pi}{10} \times 10 \times 1} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 8 \times 2} + e^{-j\frac{2\pi}{10} \times 10 \times 2} \right] + \\
 & \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 8 \times 3} + e^{-j\frac{2\pi}{10} \times 10 \times 3} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 8 \times 4} + e^{-j\frac{2\pi}{10} \times 10 \times 4} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 8 \times 5} + e^{-j\frac{2\pi}{10} \times 10 \times 5} \right] + \\
 & \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 8 \times 6} + e^{-j\frac{2\pi}{10} \times 10 \times 6} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 8 \times 7} + e^{-j\frac{2\pi}{10} \times 10 \times 7} \right] + \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 8 \times 8} + e^{-j\frac{2\pi}{10} \times 10 \times 8} \right] + \\
 & \frac{1}{2} \left[e^{-j\frac{2\pi}{10} \times 8 \times 9} + e^{-j\frac{2\pi}{10} \times 10 \times 9} \right]
 \end{aligned}$$

$$\begin{aligned}
 X(9) = & \frac{1}{2} [e^0 + e^0] + \frac{1}{2} \left[\cos \frac{16\pi}{10} - j \sin \frac{16\pi}{10} + \cos 2\pi - j \sin 2\pi \right] + \frac{1}{2} \left[\cos \frac{32\pi}{10} - j \sin \frac{32\pi}{10} + \right. \\
 & \cos 4\pi - j \sin 4\pi \left. \right] + \frac{1}{2} \left[\cos \frac{48\pi}{10} - j \sin \frac{48\pi}{10} + \cos 6\pi - j \sin 6\pi \right] + \\
 & \frac{1}{2} \left[\cos \frac{64\pi}{10} - j \sin \frac{64\pi}{10} + \cos 8\pi - j \sin 8\pi \right] + \frac{1}{2} \left[\cos \frac{80\pi}{10} - j \sin \frac{80\pi}{10} + \cos 10\pi \right. \\
 & - j \sin 10\pi \left. \right] + \frac{1}{2} \left[\cos \frac{96\pi}{10} - j \sin \frac{96\pi}{10} + \cos 12\pi - j \sin 12\pi \right] + \frac{1}{2} \left[\cos \frac{112\pi}{10} - \right. \\
 & j \sin \frac{112\pi}{10} + \cos 14\pi - j \sin 14\pi \left. \right] + \frac{1}{2} \left[\cos \frac{128\pi}{10} - j \sin \frac{128\pi}{10} + \cos \frac{16\pi}{10} - j \sin \frac{16\pi}{10} \right] \\
 & + \frac{1}{2} \left[\cos \frac{144\pi}{10} - j \sin \frac{144\pi}{10} + \cos 18\pi - j \sin 18\pi \right]
 \end{aligned}$$

$$X(9) = 1 + \frac{1}{2} [0.309 + j0.951 + 1 + j0] + \frac{1}{2} [j0.5877 - 0.809 + 1 - j0] + \frac{1}{2} [-0.809$$

$$\begin{aligned}
 & -j0.5877 + 1 - 0j] + \frac{1}{2}[-j0.951 + 0.309 + 1 - j0] + \frac{1}{2}[1 + -j0 + 1 - j0] + \frac{1}{2}[0.309 + \\
 & j0.951 + 1 - 0j] + \frac{1}{2}[j0.5877 - 0.809 + 1 - 0j] + \frac{1}{2}[-0.809 - j0.5877 - j0 + 1] + \\
 & \frac{1}{2}[-j0.951 + 0.309 + 1 - 0j]
 \end{aligned}$$

$$X(9) = 5$$

$$X(k) = \{0, 5, 0, 0, 0, 0, 0, 0, 0, 5\}$$

b) The DFT breaks down a signal into sinusoidal components in order to represent it in frequency domain. The DFT of $x(n)$ will contain two spikes (positive and negative frequencies) because it is a cosine function. By calculating projections of the signal onto complex exponentials, the DFT reveals frequency components and their amplitudes, so extracting spectral information.

$$\begin{aligned}
 2) a) \quad y(n) &= \begin{cases} \sum_{l=-\infty}^{\infty} x(n+lM) & ; 0 \leq n \leq M-1 \\ 0 & ; \text{elsewhere} \end{cases}
 \end{aligned}$$

Using DFT property for $y(n)$

$$Y(k) = \sum_{n=0}^{N-1} y(n) e^{-j\frac{2\pi}{M}kn}$$

$$\therefore y(n) = \sum_{l=-\infty}^{\infty} x(n+lM)$$

So,

$$Y(k) = \sum_{n=0}^{N-1} \sum_{l=-\infty}^{\infty} x(n+lM) e^{-j\frac{2\pi}{M}kn}$$

So, as both signals are periodic, it can be showed as:

$$Y(k) = \sum_{h=0}^{M-1} \left(\frac{2\pi k}{M} - \frac{2\pi h}{M} \right)$$

M-point DFT $Y(k)$ is a sum of shifted versions of Fourier transform $X(\omega)$ of $x(n)$.

b) The N -point DFT leverages the periodicity of the aliased sequence to analyze frequency components efficiently. In certain situations, the DFT can still identify dominant frequencies and ~~assist~~^{help} in reconstructing the original signal even while aliasing folds the spectrum.